High-Yield Pathology & Pharmacology

Chapter 1: Hematopathology - Chronic Myeloid Leukemia (CML)

Chronic Myeloid Leukemia (CML) is a myeloproliferative neoplasm characterized by the dysregulated production and uncontrolled proliferation of mature and maturing granulocytes with fairly normal differentiation. The hallmark of CML is the Philadelphia chromosome, a reciprocal translocation between chromosomes 9 and 22, designated as t(9;22)(q34;q11). This translocation fuses the BCR gene on chromosome 22 with the ABL1 gene on chromosome 9, creating the BCR-ABL1 fusion gene, which encodes a constitutively active tyrosine kinase.

Clinical Presentation: Patients often present with fatigue, weight loss, night sweats, and massive splenomegaly due to extramedullary hematopoiesis. The peripheral blood smear shows a striking leukocytosis with a full spectrum of myeloid precursors (myelocytes, metamyelocytes, bands). Unlike leukemoid reactions, leukocyte alkaline phosphatase (LAP) score is characteristically low in CML.

Treatment: The advent of targeted therapy has revolutionized the treatment of CML. Tyrosine kinase inhibitors (TKIs), such as Imatinib, specifically inhibit the BCR-ABL1 kinase activity and induce deep molecular remissions.

Chapter 2: Cardiovascular Pharmacology - Heart Failure

The management of systolic heart failure (Heart Failure with Reduced Ejection Fraction, HFrEF) relies heavily on neurohormonal blockade to prevent adverse cardiac remodeling and improve survival. Angiotensin-Converting Enzyme (ACE) Inhibitors are considered first-line agents. They inhibit the conversion of Angiotensin I to Angiotensin II, leading to vasodilation, decreased sympathetic activity, and reduced aldosterone secretion.

Important Contraindications and Adverse Effects: While ACE inhibitors are life-saving, they have strict contraindications. They are absolutely contraindicated in patients with Bilateral Renal Artery Stenosis. In these patients, GFR is highly dependent on Angiotensin II-mediated efferent arteriolar vasoconstriction. Initiating an ACE inhibitor removes this compensatory mechanism, causing an acute and profound fall in GFR and acute kidney injury.

Another major contraindication is a history of ACE-inhibitor-induced Angioedema. ACE is also responsible for the breakdown of bradykinin. Inhibition of ACE leads to bradykinin accumulation, which can cause life-threatening airway edema. If a patient develops angioedema on an ACE inhibitor, they must be switched to an Angiotensin Receptor Blocker (ARB) with caution, or a completely different class.

Other contraindications include pregnancy (teratogenic effects) and severe hyperkalemia.